

Dear Stack team,

I'm applying to the Perception Architecture team because the role sounds unusually close to the problems I've spent my career solving: cross-cutting perception architecture, novel ML approaches, real-time constraints, and turning experiments into production systems.

I co-founded Cambrian and led its technical work across computer vision, ML, GPU computing, and real-time systems. Our mixed-reality platform maintained a continuously evolving representation of the world while perception components ran at radically different cadences. Segmentation and surface-normal CNNs could take 1–3 seconds on the mobile hardware of the time, so learned affinities, tracking, geometry, optical flow, camera pose, RANSAC planes, and multi-frame evidence kept the scene coherent while delayed inference revised it as results arrived.

Underneath that was a priority-scheduled, backpressured C++ perception runtime designed around limited compute. We also built synthetic-data pipelines, model compression and quantization, segmentation and surface-normal networks, camera calibration, and cross-platform inference systems. The technology shipped to millions of users and later contributed to automotive computer-vision work, including fast C++ camera-pose estimation for a mixed-reality HUD project.

What caught my attention about Stack is the emphasis on perception problems that are not already solved in the literature. That has been much of my career: understand the underlying problem, experiment with unconventional approaches when necessary, then own enough of the stack to make the result work in production.

My current work extends that into modern AI systems. Continuum is a large Rust-based architecture for persistent, multimodal AI running across constrained consumer hardware, and Sentinel-AI explores transformer architectures that change through experience. That work has taken me deeper into foundation models, model efficiency, inference runtimes, and distributed AI while keeping me close to the systems problems I enjoy most.

I haven't spent my career inside the AV industry, and I don't claim experience I don't have with production lidar/radar stacks. What I do bring is twenty years of computer vision, perception architecture, ML research, real-time C++, constrained inference, and taking unusual ideas all the way into shipped systems. Stack's Perception Architecture team looks like a place where that combination could be useful.

Joel Teply github.com/CambrianTech